NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Lab Task(s) Report – Lab 10

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Syllabus:

Database Systems Lab

Syllabus Code: CL-2005

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Instructor:

Sir Osama Tahir

Q1)

SQL Query Text:

```
SET SERVEROUTPUT ON;

DECLARE
   v_base_salary HR.Employees.Salary%type := 250;
   v_sp_allowance_amt HR.Employees.Salary%type;
   v_f_tot_sal HR.Employees.Salary%type;

BEGIN

   v_sp_allowance_amt := v_base_salary * 0.15;
   v_f_tot_sal := v_base_salary + v_sp_allowance_amt;

   DBMS_OUTPUT.PUT_LINE('Base Sal: $' || v_base_salary);
   DBMS_OUTPUT.PUT_LINE('Special Allowance (15%): $' || v_sp_allowance_amt);
   DBMS_OUTPUT.PUT_LINE('Final Salary: $' || v_f_tot_sal);

END;
/
```

Screenshot:

Q2)

SQL Query Text:

```
SET SERVEROUTPUT ON;

DECLARE
  v_fn HR.employees.first_name%type;
  v_sal HR.employees.salary%type;

BEGIN

  SELECT first_name, salary
  into v_fn, v_sal
  from HR.employees
  where employee_id = 105;

  dbms_output.put_line('Employee First Name: ' || v_fn);
  dbms_output.put_line('Employee Salary: ' || v_sal);

END;
/
```

Screenshot:

Q3)

SQL Query Text:

```
SET SERVEROUTPUT ON;

DECLARE
   eid Number := 110;
   v_sal HR.employees.salary%type;

BEGIN

   select salary
   into v_sal
   from hr.employees
   where employee_id = eid;

   if (v_sal < 8000) then
      v_sal := v_sal + 1200;
      dbms_output.put_line('Salary was <$8000, Increased by $1200.');
   else
      v_sal := v_sal + 400;
      dbms_output.put_line('Standardized raise applied. Increased by $400.');
   end if;

   update hr.employees
   set salary = v_sal
   where employee_id = eid;

   COMMIT;
END;
/
```

Screenshot:

```
SET SERVEROUTPUT ON;

DECLARE
    eid Number := 110;
    v_sal HR.employees.salary%type;

BEGIN

    select salary
    into v_sal
    from hr.employees
    where employee_id = eid;

    if (v_sal < 8000) then
        v_sal := v_sal + 1200;
        dbms_output.put_line('Salary was <$8000, Increased by $1200.');
    else
        v_sal := v_sal + 400;
        dbms_output.put_line('Standardized raise applied. Increased by $400.');
    end if;

    update hr.employees
    set salary = v_sal
    where employee_id = eid;

    COMMIT;
END;
/
```

✓ 0.1s

```
PL/SQL procedure successfully completed.
```

Q4)

SQL Query Text:

```
SET SERVEROUTPUT ON;

DECLARE
   eid NUmber := 115;
   did HR.EMPLOYEES.DEPARTMENT_ID%TYPE;
   v_label VARCHAR2(50);

BEGIN

   select department_id
   into did
   from hr.employees
   where employee_id = eid;

   v_label := CASE did
      WHEN 90 then 'Administration'
      WHEN 60 then 'Technical Services'
      ELSE 'General Staff'
   END;

   dbms_output.put_line('Employee ID: ' || eid || ' labelled as: ' || v_label);

END;
/
```

Screenshot:

```
SET SERVEROUTPUT ON;

DECLARE
    eid NUmber := 115;
    did HR.EMPLOYEES.DEPARTMENT_ID%TYPE;
    v_label VARCHAR2(50);

BEGIN

    select department_id
    into did
    from hr.employees
    where employee_id = eid;

    v_label := CASE did
        WHEN 90 then 'Administration'
        WHEN 60 then 'Technical Services'
        ELSE 'General Staff'
    END;

    dbms_output.put_line('Employee ID: ' || eid || ' labelled as: ' || v_label);

END;
/
```

0.1s

```
PL/SQL procedure successfully completed.
```

Q5)

SQL Query Text:

```
SET SERVEROUTPUT ON;

DECLARE
   v_did NUMBER := 90;
   v_bonus NUMBER(10, 2);

BEGIN
   dbms_output.put_line('Bonus Report for Department ID: 90.');

   FOR emp IN (
      SELECT first_name, last_name, salary, NVL(commission_pct, 0) AS c_pct
      from hr.EMPLOYEES
      where department_id = v_did

   ) LOOP

      v_bonus := 0.00;

      if (emp.salary BETWEEN 15000 AND 20000) THEN
         v_bonus := (emp.salary * emp.c_pct);
         dbms_output.put_line(emp.first_name || ' ' || emp.last_name || ' | Base: $' || emp.salary || ' | Standard: $' ||
v_bonus);

      elsif (emp.salary > 20000) THEN
         v_bonus := (emp.salary * (emp.c_pct + 0.25));
         dbms_output.put_line(emp.first_name || ' ' || emp.last_name || ' | Base: $' || emp.salary || ' | Standard: $' ||
v_bonus);

      ELSE
         dbms_output.put_line(emp.first_name || ' ' || emp.last_name || ' | Base: $' || emp.salary || 'No Bonus
Applicable.');

      end if;

   end loop;

EXCEPTION
   WHEN NO_DATA_FOUND THEN
      DBMS_OUTPUT.PUT_LINE('No employee is in the department ID: 90.');

END;
/
```

Screenshot:

```
DECLARE
    v_did NUMBER := 90;
    v_bonus NUMBER(10, 2);

BEGIN

    dbms_output.put_line('Bonus Report for Department ID: 90.');

    FOR emp IN (
        SELECT first_name, last_name, salary, NVL(commission_pct, 0) AS c_pct
        from hr.EMPLOYEES
        where department_id = v_did

    ) LOOP

        v_bonus := 0.00;

        if (emp.salary BETWEEN 15000 AND 20000) THEN
            v_bonus := (emp.salary * emp.c_pct);
            dbms_output.put_line(emp.first_name || ' ' || emp.last_name || ' | Base: $' || emp.salary || ' | Standard: $' || v_bonus);

        elsif (emp.salary > 20000) THEN
            v_bonus := (emp.salary * (emp.c_pct + 0.25));
            dbms_output.put_line(emp.first_name || ' ' || emp.last_name || ' | Base: $' || emp.salary || ' | Standard: $' || v_bonus);

        ELSE
            dbms_output.put_line(emp.first_name || ' ' || emp.last_name || ' | Base: $' || emp.salary || 'No Bonus Applicable.');

        end if;

    end loop;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('No employee is in the department ID: 90.');

END;
/
```

✓ 0.0s

```
PL/SQL procedure successfully completed.
```

Q6)

SQL Query Text:

```
BEGIN
   DBMS_OUTPUT.PUT_LINE('--- Shipping Department Employees ---');
   FOR emp_rec IN (
      SELECT first_name, last_name, salary
      FROM hr.employees
      WHERE department_id = 50
   )
   LOOP
      DBMS_OUTPUT.PUT_LINE('Name: ' || emp_rec.first_name || ' ' || emp_rec.last_name || ' | Salary: $' ||
emp_rec.salary);
   END LOOP;
END;
/
```

Screenshot:

Q7)

SQL Query Text:

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE display_emp_salary (
    p_emp_id IN NUMBER
) IS
    v_salary hr.employees.salary%TYPE;

BEGIN
    SELECT salary INTO v_salary
    FROM hr.employees
    WHERE employee_id = p_emp_id;

    DBMS_OUTPUT.PUT_LINE('The current salary for employee ID ' || p_emp_id || ' is $' || v_salary);
END;
/

EXEC display_emp_salary(105);;
```

Screenshot:

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE display_emp_salary (
    p_emp_id IN NUMBER
) IS
    v_salary hr.employees.salary%TYPE;

BEGIN
    SELECT salary INTO v_salary
    FROM hr.employees
    WHERE employee_id = p_emp_id;

    DBMS_OUTPUT.PUT_LINE('The current salary for employee ID ' || p_emp_id || ' is $' || v_salary);
END;
/

EXEC display_emp_salary(105);
```

Script Output | Task completed in 0.05 seconds

```
The current salary for employee ID 105 is $5280


PL/SQL procedure successfully completed.


Procedure DISPLAY_EMP_SALARY compiled

The current salary for employee ID 105 is $5280
```

Dbms Output | Buffer Size: 20000

homeuser

```
The current salary for employee ID 105 is $5280
```

Q8)

SQL Query Text:

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE get_emp_dept (
   p_emp_id IN NUMBER,
   p_dept_id OUT NUMBER
) IS
BEGIN
   SELECT department_id INTO p_dept_id
   FROM hr.employees
   WHERE employee_id = p_emp_id;
END;
/
DECLARE
   v_dept_id NUMBER;
BEGIN
   get_emp_dept(105, v_dept_id);
   DBMS_OUTPUT.PUT_LINE('The retrieved Department ID is: ' || v_dept_id);
END;
/
```

Screenshot:

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE get_emp_dept (
    p_emp_id IN NUMBER,
    p_dept_id OUT NUMBER
) IS
BEGIN
    SELECT department_id INTO p_dept_id
    FROM hr.employees
    WHERE employee_id = p_emp_id;
END;
/
DECLARE
    v_dept_id NUMBER;
BEGIN
    get_emp_dept(105, v_dept_id);
    DBMS_OUTPUT.PUT_LINE('The retrieved Department ID is: ' || v_dept_id);
END;
/
```

Script Output

Task completed in 0.103 seconds

```
Procedure GET_EMP_DEPT compiled

The retrieved Department ID is: 60


PL/SQL procedure successfully completed.
```

Dbms Output

Buffer Size: 20000

homeuser

```
The retrieved Department ID is: 60
```

Q9)

SQL Query Text:

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE increase_by_25_percent (
    p_value IN OUT NUMBER
) IS
BEGIN
    p_value := p_value * 1.25;
END;
/

DECLARE
    v_my_number NUMBER := 200;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Original Value: ' || v_my_number);
    increase_by_25_percent(v_my_number);
    DBMS_OUTPUT.PUT_LINE('Updated Value (After Procedure): ' || v_my_number);
END;
/
```

Screenshot:

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE increase_by_25_percent (
    p_value IN OUT NUMBER
) IS
BEGIN
    p_value := p_value * 1.25;
END;
/

DECLARE
    v_my_number NUMBER := 200;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Original Value: ' || v_my_number);
    increase_by_25_percent(v_my_number);
    DBMS_OUTPUT.PUT_LINE('Updated Value (After Procedure): ' || v_my_number);
END;
/
```

Script Output

Task completed in 0.054 seconds

```
Procedure INCREASE_BY_25_PERCENT compiled

Original Value: 200
Updated Value (After Procedure): 250


PL/SQL procedure successfully completed.
```

Dbms Output

Buffer Size: 20000

homeuser

```
Original Value: 200
Updated Value (After Procedure): 250
```

Q10)

SQL Query Text:

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE add_new_location (
    p_street_address IN VARCHAR2,
    p_city IN VARCHAR2
) IS
    v_next_loc_id hr.locations.location_id%TYPE;
BEGIN
    SELECT NVL(MAX(location_id), 0) + 100
    INTO v_next_loc_id
    FROM hr.locations;

    INSERT INTO hr.locations (location_id, street_address, city)
    VALUES (v_next_loc_id, p_street_address, p_city);

    DBMS_OUTPUT.PUT_LINE('Success: New location added in ' || p_city || ' with Location ID: ' ||
v_next_loc_id);
    COMMIT;
END;
/
EXEC add_new_location('123 Sunset Boulevard', 'Karachi');
```

Screenshot:

Worksheet | Query Builder

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE add_new_location (
    p_street_address IN VARCHAR2,
    p_city IN VARCHAR2
) IS
    v_next_loc_id hr.locations.location_id%TYPE;
BEGIN
    SELECT NVL(MAX(location_id), 0) + 100
    INTO v_next_loc_id
    FROM hr.locations;

    INSERT INTO hr.locations (location_id, street_address, city)
    VALUES (v_next_loc_id, p_street_address, p_city);

    DBMS_OUTPUT.PUT_LINE('Success: New location added in ' || p_city || ' with Location ID: ' || v_next_loc_id);
    COMMIT;
END;
/
EXEC add_new_location('123 Sunset Boulevard', 'Karachi');
```

Script Output | Task completed in 0.047 seconds

```
Procedure ADD_NEW_LOCATION compiled

Success: New location added in Karachi with Location ID: 3301


PL/SQL procedure successfully completed.
```

Dbms Output | Buffer Size: 20000

homeuser

```
Success: New location added in Karachi with Location ID: 3301
```